

Tools for Applied Monitoring, Evaluation and Learning

Beyond Analysis: Strategic Evaluation with Stata

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Eric Booth Elizabeth Teas

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Preface: The Modern Evaluator's Dilemma

Applied evaluation has moved beyond simple pre-post t-tests. Today's evaluator must be part data engineer, part AI prompt engineer, and part visualization specialist. This book provides a roadmap for using Stata as the central "brain" of this complex ecosystem.

**PART I: THE EMBEDDED
EVALUATOR
PRINCIPLES, SPEED, AND THE
MODERN STACK**

Strategic Evaluation in Implementation Science

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1.1 The Practitioner-Researcher Relationship

Explores the "Embedded Evaluation" model. We discuss moving away from "parachute research" toward sustained relationships where the evaluator is a strategic partner in the implementation team.

- ▶ **Bridging the gap:** Shared goals vs. analytical independence. How to maintain rigor while being "in the room" where decisions are made.
- ▶ **Feedback loops:** Using Stata to provide real-time insights for program adaptation.
- ▶ **Messy Contexts:** Working with nonprofits where data collection is often secondary to service delivery.

1.2 Implementation Science Principles in Practice

This section defines the core IS frameworks (e.g., RE-AIM, CFIR) and how they translate to data structures in Stata.

Implementation science focuses on the 'how' and 'why' of program success, not just the 'what'. See [James2013] for foundational concepts.

- ▶ **Fidelity Monitoring:** Tracking if the program is being delivered as intended.
- ▶ **Adaptation Tracking:** Coding and managing changes made to the intervention in real-time.
- ▶ **Determinants of Change:** Identifying barriers and facilitators through nested data structures.

1.3 Stata as the Central Evaluation Brain

Why Stata is the ideal bridge: its ability to handle messy, real-world data while providing the rigor required for publication and the speed required for stakeholders.

- ▶ **Reproducibility:** Why do-files are superior to point-and-click dashboards for audit trails.
- ▶ **Extensibility:** Using community-contributed commands to solve thorny evaluation problems.

Setting Up a High-Performance Environment

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2.1 The Speed Revolution: ftools and gtools

Detailed guide on using faster alternatives to core commands.

- ▶ **Benchmarking:** Comparing `gcollapse` vs. `collapse` on large administrative datasets.
- ▶ **Multi-way Fixed Effects:** Using `reghdfe` for complex nested evaluation designs.
- ▶ **Efficient Hashing:** Understanding why these tools work and when to use them.

When working with millions of administrative records, standard Stata commands can become bottlenecks. We implement `ftools` and `gtools` to reclaim your time.

2.2 Monitoring Real-Time Participation

Techniques for building "Command Centers" in Stata to track response rates and program reach.

- ▶ **Response Rate Trackers:** Automating daily reports from Qualtrics/REDCap APIs.
- ▶ **Attrition Alerts:** Using Stata to flag participants at risk of dropping out before they do.
- ▶ **Geographic Coverage:** Quick mapping to see where participants are (and aren't) coming from.

2.3 Communicating with Funders and Stakeholders

Using Stata to automate the "Monday Morning Update" for nonprofit boards and foundation officers.

- ▶ **Dynamic Tables:** Moving beyond copy-pasting to automated `putexcel` and collect workflows.
- ▶ **Funder-Ready Outputs:** Designing visuals that tell a story of impact and reach without the jargon.

2.4 Integrating AI: The Gemini CLI Bridge

Setup guide for integrating Node.js and the Gemini CLI with Stata.

- ▶ **Authentication:** Workflows for macOS, Windows, and Linux.
- ▶ **Prompting from Do-files:** Running AI-assisted cleaning routines directly within your pipeline.
- ▶ **Privacy:** Strategies for de-identifying data before sending it to an LLM.

The `gemini` package allows Stata to communicate directly with Google's LLMs. We'll cover API key management and shell security.

PART II: DATA ENGINEERING MODERNIZATION, CLEANING, AND HARMONIZATION

3.1 Web Scraping and API Pipelines

Walks through examples of scraping program data and connecting to APIs like REDCap, Qualtrics, and Census Bureau feeds using Stata.

3.2 Unstructured Data: The AI Bridge

Using LLMs within Stata to turn qualitative interview fragments or messy program notes into structured, analyzable variables using the `gemini` package.

Harmonization and Master Data Management

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4.1 Merging the Unmergable

Fuzzy matching techniques and crosswalk building. Dealing with shifting geographic IDs and changing program taxonomies over time.

4.2 Longitudinal Panel Construction

Building "Long" files from "Wide" chaos. Techniques for tracking program participants across multiple years of intervention in implementation settings.

5.1 The Privacy-Utility Tradeoff

Techniques for de-identification and the creation of "Research-Ready" public-use files for non-profit and public sector clients.

Data ethics isn't just a compliance box—it's about building trust with participants.

5.2 Synthetic Data Generation

Using Stata's `simulate` and AI-driven synthesis to create realistic mock datasets that preserve the covariance matrix of the original sensitive data.

**PART III: THE NEW CAUSAL
FRONTIER
INNOVATIVE AND
NON-STANDARD ANALYSIS**

Modern Difference-in-Differences

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6.1 The Revolution in Staggered Adoption

Explaining the recent shift in DiD methodology and why Two-Way Fixed Effects (TWFE) often fails in applied evaluation.

6.2 Implementing `csdid` and `jwdid`

A step-by-step guide to estimating heterogeneous treatment effects and producing event-study plots that survive academic and client scrutiny.

Synthetic Controls and Matrix Completion

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7.1 The Synthetic Control Method (**synth**)

Building a "counterfactual" unit for single-treated unit evaluations.

7.2 Synthetic Difference-in-Differences (**sdid**)

Combining the strengths of DiD and Synthetic Control for robust program estimation in complex implementation contexts.

8.1 Targeting Models with Lasso and Elastic Net

Using `lassopack` to identify which program participants are most likely to drop out or succeed, enabling proactive implementation adjustments.

8.2 Random Forests in Stata

Applying non-linear classification to predict program outcomes and visualize feature importance for stakeholders.

**PART IV: HIGH-IMPACT
COMMUNICATION
SPARKLINES, WEBDOCS, AND
INTERACTIVE REPORTING**

9.1 Sparklines and Micro-charts with sparkta

Using sparkta to create dense, data-rich dashboards that show trends at a glance across hundreds of units.

Fahad Mirza's sparkta package allows us to generate thousands of "sparklines" (micro-charts) for high-density dashboards.

9.2 Advanced Coefficient Plots

Visualizing effect sizes and confidence intervals across multiple models for easy stakeholder comparison.

Dynamic Reporting and Web Integration

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10.1 Dynamic HTML with webdoc

Building live, searchable project websites directly from your do-files to keep practitioner-partners informed.

By embedding sparkta outputs directly into webdoc workflows, we can generate auto-updating HTML project portals.

10.2 AI-Assisted Narrative Generation

Using the gemini package to draft the "Executive Summary" based on your Stata output tables.

**PART V: BUILDING TOOLS
THAT Mata-ER
CUSTOM DEVELOPMENT AND
SCALABILITY**

Building Internal Evaluation Toolkits

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11.1 From Scripts to Systems

Turning project-specific cleaning code into standardized organization-wide ado-files.

11.2 Writing Help Files (.sthlp)

Best practices for documenting institutional knowledge and custom analytical processes for team onboarding.

Scalability and Big Data in the Public Sector

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12.1 Introduction to Mata for Evaluators

When even `gtools` isn't fast enough. Matrix manipulation and custom functions for large-scale administrative data.

APPENDICES

Appendix A: The Gemini-Stata Setup Guide

Appendix B: The Modern Causal Toolbox

